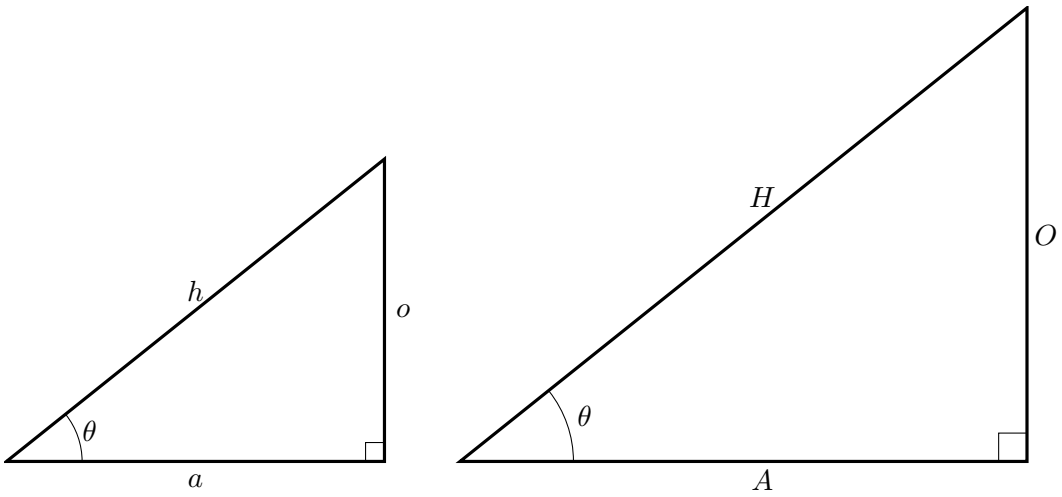
A Crash Course in Trigonometry

Right-Angled Triangles

Consider two right-angled triangles with one identical angle (other than the right angle):



Questions

- What can you say about the third angle in each of the triangles?
- Therefore, the two triangles are _____.
- This implies that:

$$\frac{o}{h} = ___, \text{ and } \quad \frac{a}{h} = ___, \quad \frac{o}{a} = ___.$$

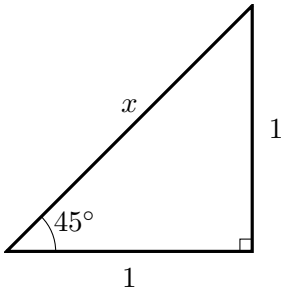
Conclusion: The above ratios only depend on _____.

> **Definitions - Basic Trig Functions:** given an angle θ in a right-angled triangle, we define the following three functions:
>
> $$\sin\theta = \frac{\text{opp}}{\text{hyp}}, \ \cos\theta = \frac{\text{adj}}{\text{hyp}}, \ \tan\theta = \frac{\text{opp}}{\text{adj}},$$
>
> where opp, and adj are the lengths of the sides of the corresponding right-angled triangle positions opposite, and adjacent to angle θ respectively, and hyp is the length of the hypotenuse.

Question For what angles are these functions currently defined? i.e. What are their domains?

Special Values of the Trig Functions

Question By finding the value of x in the following $45° - 45° - 90°$ triangle exactly (no decimals!), compute the values below:

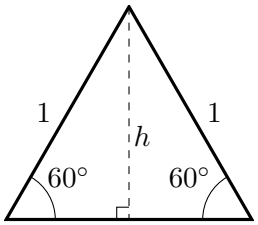


$\sin 45° =$ _____

$\cos 45° =$ _____

$\tan 45° =$ _____

Question By finding the value of h (i.e. the length of the dashed line) in the following $60° - 60° - 60°$ triangle exactly (no decimals!), compute the values below:



$\sin 60° =$ _____

$\cos 60° =$ _____

$\tan 60° =$ _____

You can also use the same triangle to compute the following values (try rotating the page on its side):

$$\sin 30° = _____,\ \cos 30° = _____, \tan 30° = _____.$$

Question By imagining what happens to the lengths of each of the sides of a right-angled triangle as the angle θ approaches $0°$, find the following values:

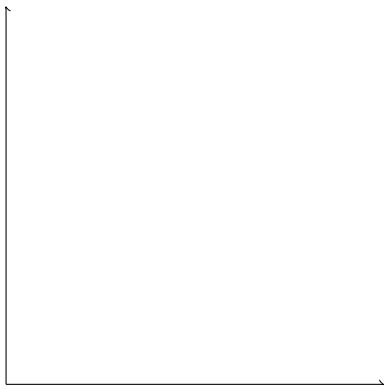
$$\sin 0° = _____,\ \cos 0° = _____, \tan 0° = _____.$$

Question By imagining what happens to the lengths of each of the sides of a right-angled triangle as the angle θ approaches $90°$, find the following values:

$$\sin 90° = _____,\ \cos 90° = _____, \tan 90° = _____.$$

The Unit Circle

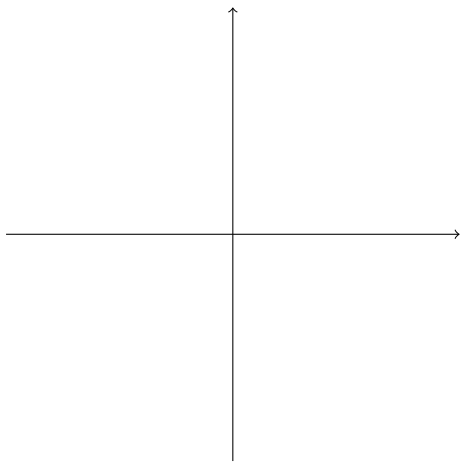
Suppose that the hypotenuse of a right-angled triangle has length 1. Draw such a triangle on the axes to the right, with its angle θ located at the origin, and the adjacent edge on the x-axis. Imagine the angle increasing from $0°$ to $90°$. What shape does the end of the hypotenuse trace out? Draw this shape on the axes.



Now fix the angle θ, and label the corresponding point on your traced shape (x, y). Then

$$\cos\theta = ___, \text{ and } \sin\theta = ___.$$

Continue drawing your shape all the way around on the next set of axes. Label an angle with $90° < \theta < 180°$.



For such an angle, we *define* $\cos\theta = x$, and $\sin\theta = y$, where x and y are the coordinates of the point on the unit circle corresponding to the angle θ, as measured anti-clockwise from the positive horizontal axis.

Question For angles $90° < \theta < 180°$, is $\sin\theta$ positive or negative? What about $\cos\theta$?

Definitions - Trig Functions for General Angles

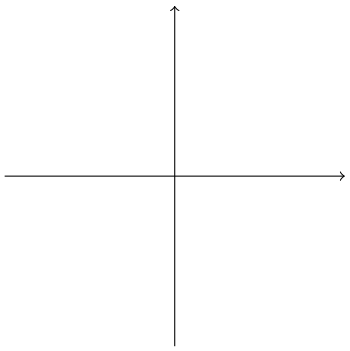
Given any angle θ, $\sin\theta$ is the y-coordinate of the point on the unit circle whose corresponding radius makes the angle θ with the positive horizontal axis, measure anti-clockwise. $\cos\theta$ is the x-coordinate of the same point. To get negative angles, measure clockwise instead.

For an angle θ, and the corresponding point on the unit circle (x, y),

$$\tan\theta = _____ = _____.$$

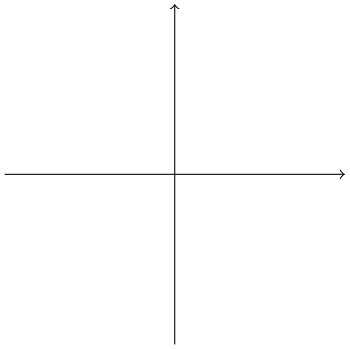
Reference Triangles

For an angle θ in each of the second, third, or fourth quadrants, there is a corresponding angle in the first quadrant, defining a *reference triangle*. You can use the latter to calculate values of each of the trig functions for the angle θ.



Question On each of the three sets of axes to the right, draw a unit circle and an angle in quadrant 2, 3, and 4 respectively. Show the corresponding reference triangles in the first quadrant.

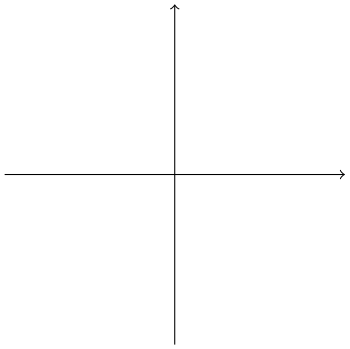
Question Using these, decide whether each of the three trig functions takes positive values and which takes negative value in each of the four quadrants. In the following table, put '+' or '-' in each of the nine spaces.



	$\sin\theta$	$\cos\theta$	$\tan\theta$
2nd quad			
3rd quad			
4th quad			

More Special Values of the Trig Functions

Using your known values of sin, cos, and tan for the angle 0°, 30°, 45°, 60°, and 90° and the idea of reference triangles above, find the values of each of the three functions for all of the following angles: 120°, 135°, 150°, 180°, 210°, 225°, 240°, 270°, 300°, 315° and 330°. Show your work. Fill in these angles and the corresponding blank spaces on the unit circle drawn on the last page of this packet.

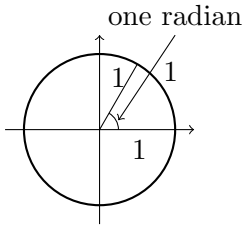


Note: The unit circle will be an extremely useful reference for you for the next few weeks. I suggest detaching it from this pack and bringing it every day with you to class.

Radians

For reasons that will become apparent next week, degrees are not a sensible measure of angles for any work with trig that involves calculus. Instead, we will measure angles in *radians*.

Definition - Radian: One *radian* is the angle in the unit circle at which the corresponding arc-length is exactly 1:



1. What is the circumference of the unit circle?

2. If the arc-length is 1, what fraction is the arc as part of the entire circle?

3. If the angle is θ°, what fraction of the whole circle is that?

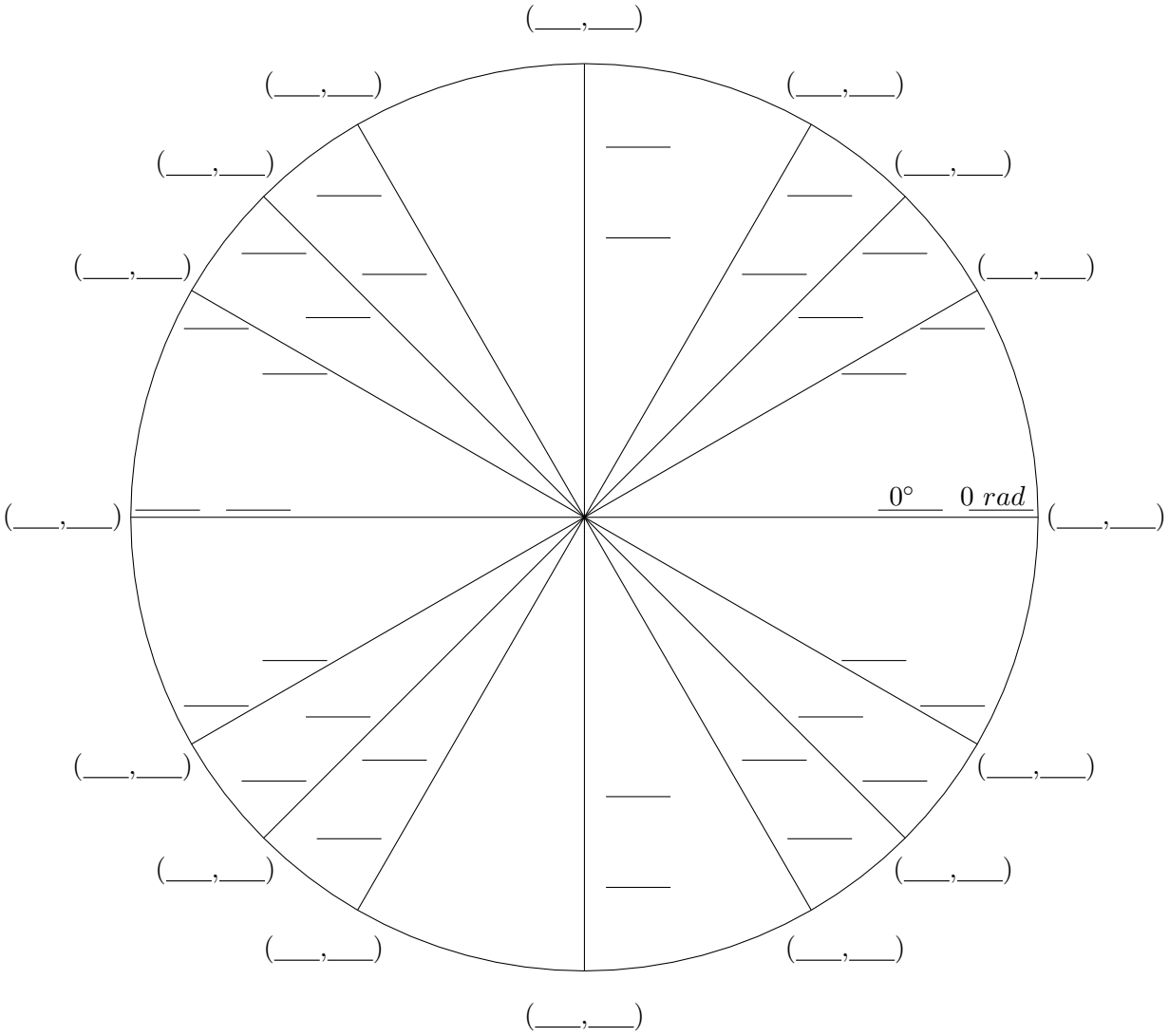
4. What can you say about your last two answers, and why?

5. Use your answer to the last question to find the size of one radian in degrees.

6. Lastly, fill on all the radian measures of the angles on the unit circle on the attached page.

- One radian is $\left(\quad\right)^\circ \approx$ _____$^\circ$.
- $1^\circ = \left(\quad\right)$ radians $\approx$ _____ radians.

The Unit Circle



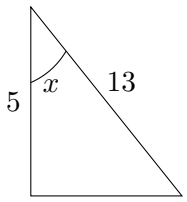
A Crash Course in Trigonometry - Homework

Solving Triangles - Trigonometry Angle Problems

1. For each value of θ below, draw the unit circle and the appropriate "reference triangle." Then use this to determine the *exact* values of $\sin\theta$ and $\cos\theta$. (Make sure to label and/or show all of your work; do NOT use your calculator.)

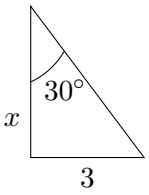
   (a) $\theta = \frac{5\pi}{6}$ (b) $\theta = -\frac{\pi}{4}$ (c) $\theta = 480^\circ$ (d) $\theta = \frac{\pi}{3} + 1,000,000\pi$

2. Given the right triangle to the right, find the *exact* values of $\sin x$, $\cos x$, and $\tan x$.



3. Suppose that the terminal side of an angle θ lies in Quadrant III and lies on the line $y = 3x$. Find $\sin\theta$.

4. Given the right triangle to the right, solve for x.



5. Given that $\tan\theta = -\frac{5}{4}$ and $\cos\theta > 0$, find the exact values of $\sin\theta$ and $\cos\theta$. Show your work clearly; your work should use a sketch of the unit circle.

6. Let t be an angle in the first quadrant, as pictured below. Evaluate the following expressions in terms of a.

   (a) $\sin(t + 2\pi) =$ _____

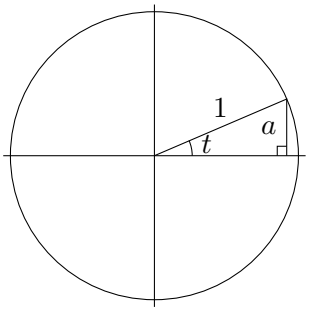
   (b) $\sin(t + \pi) =$ _____

   (c) $\cos\left(\frac{\pi}{2} - t\right) =$ _____

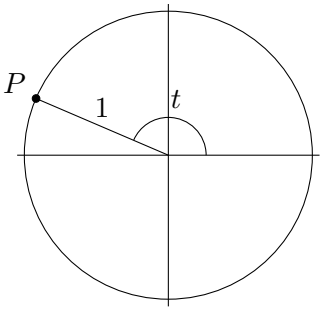
   (d) $\sin(\pi - t) =$ _____

   (e) $\sin(2\pi - t) =$ _____

   (f) $\cos\left(\frac{3\pi}{2} - t\right) =$ _____



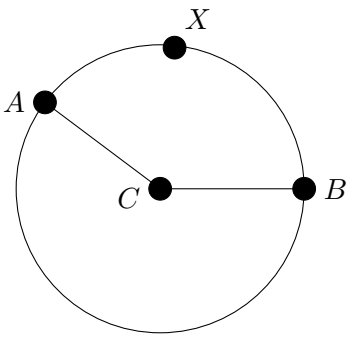
7. True or False: In the diagram to the right, the coordinates of point P are $(-\cos t, \sin t)$.



8. Find the x and y coordinates on the unit circle determined by the following angles. If possible, find these coordinates exactly. Otherwise, approximate them.

   (a) 30° (b) 315° (c) 130° (d) −240° (e) 1000°

9. Suppose the circle pictured below has radius 8 cm. Complete the following statements.



   (a) If the measure of angle ACB is 128 degrees, then the measure of angle ACB is _____ radians and the measure of arc AXB is _____ cm.

   (b) If the measure of angle ACB is 2 radians, then the measure of angle ACB is _____ degrees and the measure arc AXB is _____ cm.

   (c) If the measure of arc AXB is 15 cm, then the measure of angle ACB is _____ degrees and the measure of angle ACB is _____ radians.

10. For what values of θ is:

$$\text{(a) } \sin(\theta) \geq 0? \quad \text{(b) } \cos(\theta) \leq 0? \quad \text{(c) } \sin(\theta) \geq \cos(\theta)?$$

11. Find the exact values of the following.

$$\text{(a) } \sin\left(\tfrac{\pi}{3}\right) \quad \text{(b) } \cos\left(-\tfrac{3\pi}{4}\right) \quad \text{(c) } \sin^2(2.3) + \cos^2(2.3) \quad \text{(d) } \tan\left(\tfrac{7\pi}{6}\right)$$

Solving Triangles - Trigonometry Word Problems

1. A rocket is fired at sea level and climbs at a constant angle of 75° through a distance of 10,000 feet. Approximate its altitude to the nearest foot.

2. An airline pilot wishes to make his approach to an airstrip at an angle of 10° with the horizontal. If he is flying at an altitude of 5000 feet, approximately how far from the airstrip should he begin his descent?

3. A 16 foot long ladder is leaning against a wall and making a 60° angle with the ground. Without using your calculator determine exactly how high on the wall the top of the ladder is resting.

4. An astronomer is studying two distant stars each approximately 12 thousand light years from earth. She finds that the angle spanned by the two stars, with the earth at its vertex, is approximately 74°. Estimate the distance between the two stars.

5. A CB antenna is located on the top of a garage that is 16 feet tall. From a point on level ground that is 100 feet from a point directly below the antenna, the antenna subtends an angle of 12°. Approximate the length of the antenna.

6. From a point A that is 8 meters above level ground, the angle of elevation of the top of a building is 31° and the angle of depression of the base of the building is 12°. Approximate the height of the building.

7. A motorist, traveling along a level highway at a speed of 60 km/h directly toward a distant mountain, observes that between 1:00 p.m. and 1:10 p.m. the angle of elevation of the top of the mountain changes from 10° to 70°. Approximate the height of the mountain.

8. The center of the great clock in the bell tower of the British parliament building is approximately 250 feet above the ground. The minute hand on the clock is 14 feet long. How high above the ground is the tip of the minute hand when the clock reads ten minutes after the hour?

9. The Hubble space telescope orbits the earth every 90 minutes in a near circular orbit 370 miles above the surface of the earth. (The radius of the earth is 3960 miles.)
   (a) What distance along its orbit will the Hubble telescope travel in 1 hour?
   (b) What is the radian measure of the angle through which the Hubble telescope will move after traveling 1000 miles along its orbit? (The angle is measured from the center of the earth.)

Graphing and Transforming Trigonometric Functions

1. Use Geogebra to help you sketch the following on three sets of axes (one per question). The domain of the graphs should be $[-2\pi, 2\pi]$. Be sure to label your axes.
   (a) i. $y = \sin(x)$ ii. $y = \frac{1}{2}\sin(x)$ iii. $y = -3\sin(x)$
   (b) i. $y = \cos(x)$ ii. $y = \cos(2x)$ iii. $y = \cos\left(\frac{1}{3}x\right)$
   (c) i. $y = \sin(x)$ $y = \sin\left(x - \frac{\pi}{4}\right)$ $y = \sin\left(x + \frac{\pi}{2}\right)$

2. Match the following transformations with the description of their effects. (Assume $k > 1$.)

   (a) $f(x-k)$ (b) $f(x)+k$ (c) $f\left(\frac{1}{k}x\right)$ (d) $kf(x)$

   (e) $f(x+k)$ (f) $f(x)-k$ (g) $f(kx)$ (h) $\frac{1}{k}f(x)$

Vertical shift k units down	Horizontal stretch by a factor of k
_____	_____
Vertical shrink by a factor of k	Horizontal shift k units to the left
_____	_____
Vertical shift k units up	Horizontal shrink by a factor of k
_____	_____
Vertical stretch by a factor of k	Horizontal shift k units to the right
_____	_____

3. Sketch the following graphs by hand. The range of the graphs should be $[-2\pi, 2\pi]$. Be sure to label your axes.

   (a) $y = 3\sin(2x)$ (b) $y = -\frac{1}{2}\cos 2\left(x - \frac{\pi}{2}\right)$ (c) $y = \cos\left(\frac{1}{3}x\right) - 1$ (d) $y = |3\sin(2x)|$

4. Find possible formulas for the trigonometric functions graphed below.

